Decomposition analyses inc. abnormal savannas (AI; reordered; dredge-selected best models refit with REML)

30 April 2026

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0.1 Packages

0.2 Data (CSV, filtering, AI extraction)

Data summary

CSV_used	N	Sites	AI_min	AI_max
C:/Users/RTHOR95/OneDrive - The University of Liverpool/R/joel_termites/Termite_Decomp/Termite_Decomp/Updated_DB4_NEW.csv	35	10	0.353	1.792

0.3 Helpers (print-friendly tables, best-model refit, R², plotting style)

1 1) Termite absolute (Term_Absolute)

(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
57.41	-15.17	NA	NA	NA	NA	4	-139.36	298.29	0.00	0.28
45.32	NA	NA	NA	NA	NA	3	-144.03	299.30	1.01	0.17
73.95	-27.30	NA	+	NA	NA	5	-135.50	299.62	1.34	0.14
30.39	-17.66	1.16	NA	NA	NA	5	-137.72	300.55	2.26	0.09

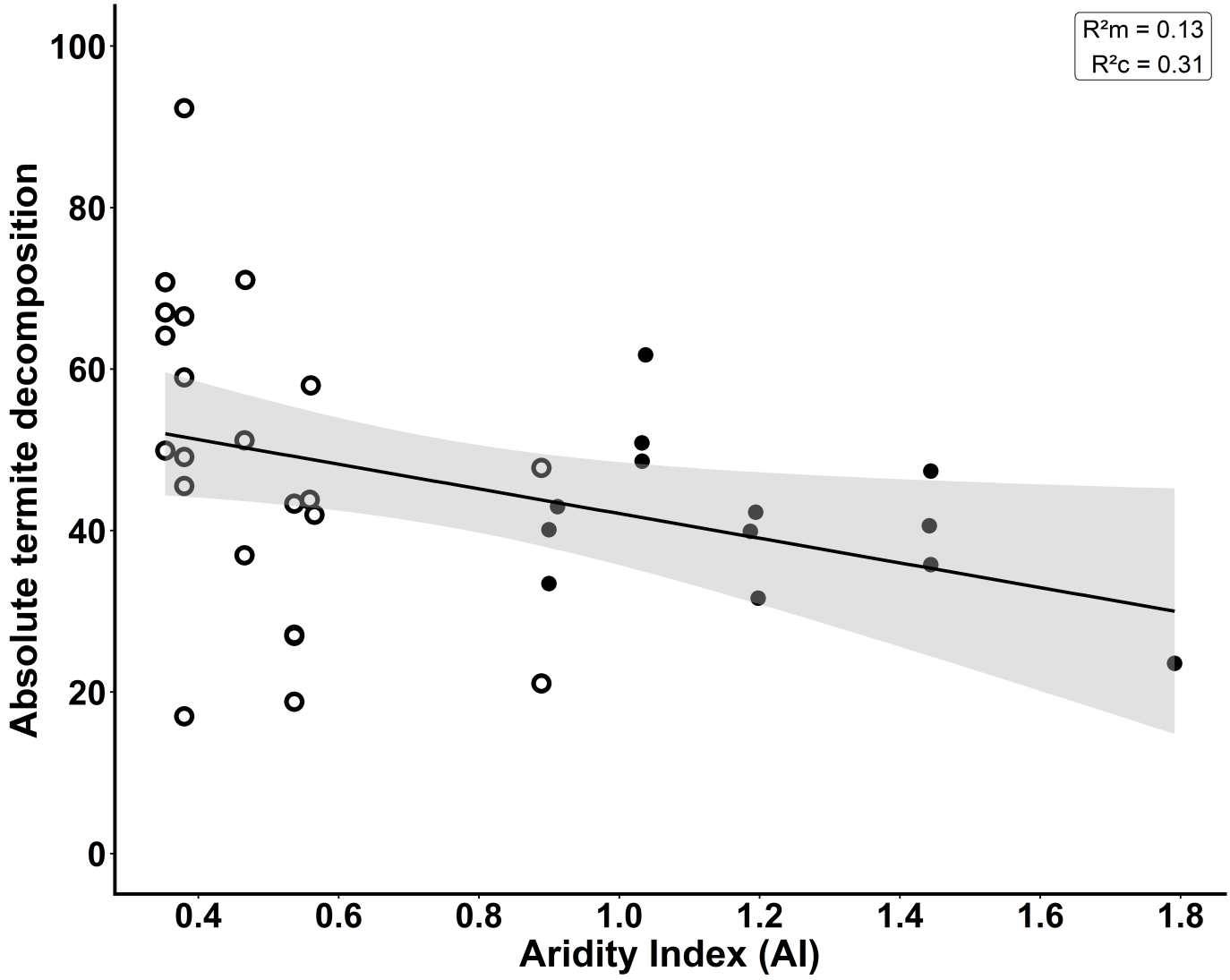
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
9.03	-15.66	1.97	+	+	NA	7	-128.41	301.02	2.73	0.07
60.24	-16.14	NA	+	+	NA	6	-130.61	301.09	2.81	0.07
42.08	NA	NA	+	NA	NA	4	-140.87	301.13	2.85	0.07
44.99	NA	0.01	NA	NA	NA	4	-142.61	302.01	3.72	0.04
47.24	-29.30	1.13	+	NA	NA	6	-133.90	302.05	3.76	0.04
31.76	NA	0.41	+	NA	NA	5	-139.39	303.99	5.70	0.02
50.18	-15.93	0.39	+	+	+	8	-125.05	305.03	6.74	0.01
134.84	-30.42	-2.20	+	NA	+	7	-130.37	305.32	7.03	0.01
-172.04	NA	8.27	+	NA	+	6	-135.71	306.80	8.51	0.00

Selected_model_formula
Term_Absolute ~ AI + (1 | Site)

nobs	AIC	BIC	logLik	sigma
35	286.73	292.95	-139.36	14.11

term	estimate	std_error	df	t_value
(Intercept)	57.41	7.21	8.56	7.96
AI	-15.17	8.08	11.12	-1.88

R2m	R2c
0.13	0.31



2 2) Microbe absolute (Microbe_Absolute)

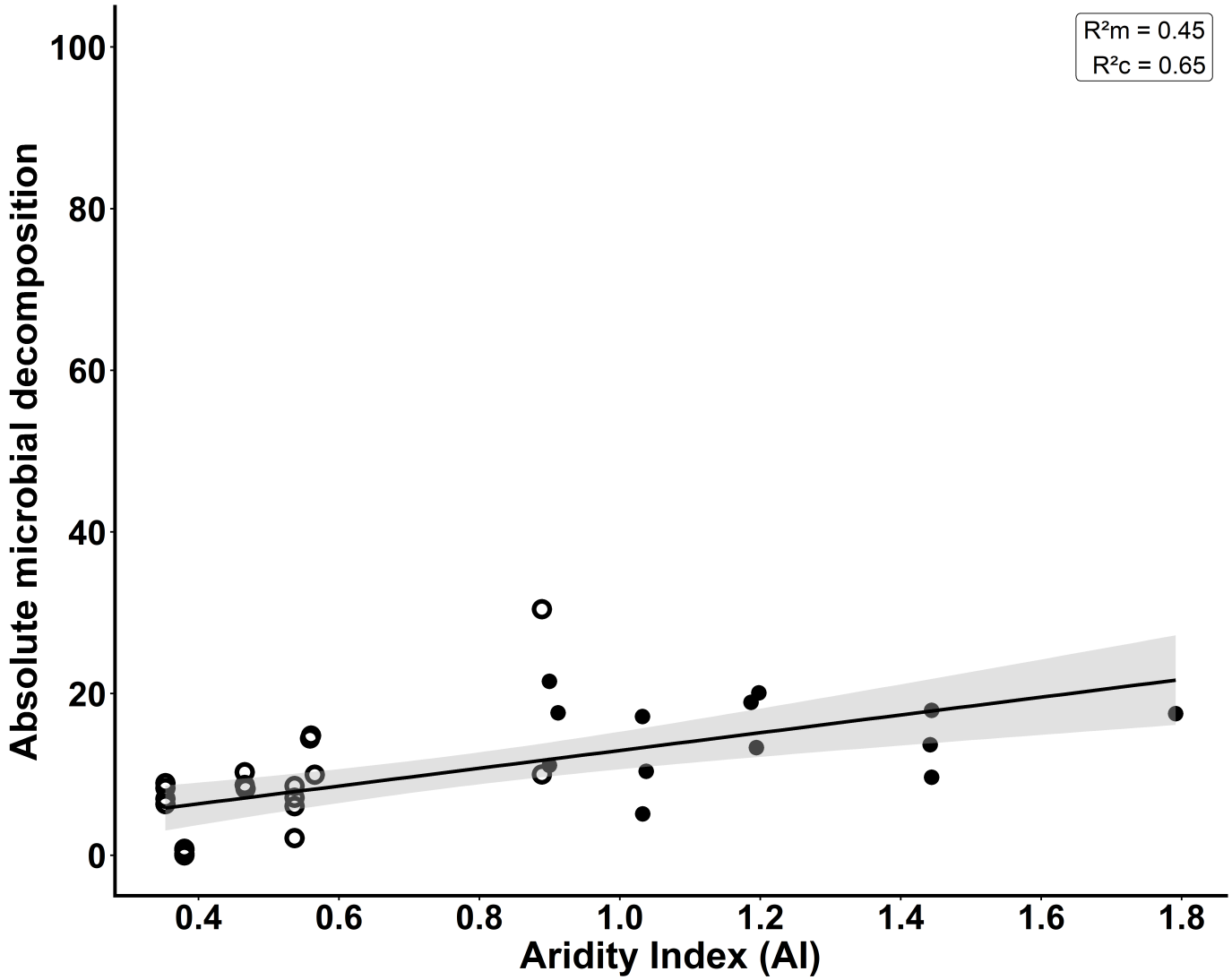
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
-23.00	7.15	1.13	NA	NA	NA	5	-100.21	219.27	0.00	0.30
3.53	9.12	NA	NA	NA	NA	4	-102.33	220.69	1.42	0.15
12.97	1.72	NA	+	+	NA	6	-94.49	220.91	1.64	0.13
-8.90	2.41	0.81	+	+	NA	7	-93.13	221.32	2.04	0.11
-26.37	9.48	1.13	+	NA	NA	6	-97.83	221.79	2.52	0.08
-28.29	NA	1.57	NA	NA	NA	4	-104.57	222.06	2.79	0.07
0.11	11.56	NA	+	NA	NA	5	-99.91	222.96	3.69	0.05
-22.72	NA	1.41	+	NA	NA	5	-102.29	223.37	4.10	0.04
11.13	NA	NA	NA	NA	NA	3	-107.44	224.74	5.47	0.02
1.45	2.24	0.42	+	+	+	8	-90.73	225.11	5.84	0.02
-53.40	10.06	2.15	+	NA	+	7	-95.34	225.14	5.87	0.02
60.22	NA	-1.80	+	NA	+	6	-99.62	226.06	6.79	0.01
12.83	NA	NA	+	NA	NA	4	-105.06	226.06	6.79	0.01

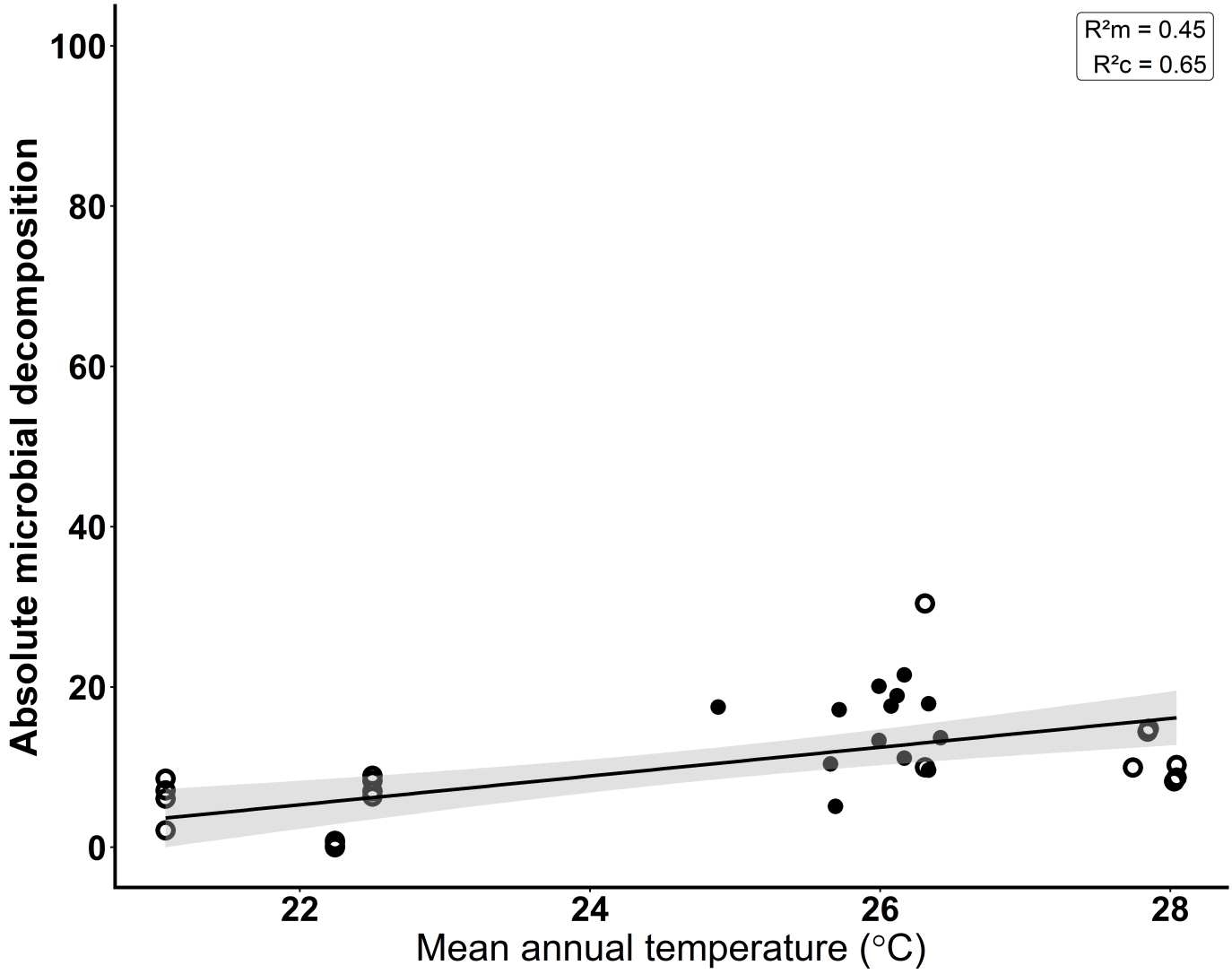
Selected_model_formula
Microbe_Absolute ~ AI + Avg_Temp + (1 | Site)

nobs	AIC	BIC	logLik	sigma
35	210.42	218.2	-100.21	4.21

term	estimate	std_error	df	t_value
(Intercept)	-23.00	13.51	7.58	-1.70
AI	7.15	3.08	11.20	2.32
Avg_Temp	1.13	0.57	8.09	1.99

R2m	R2c
0.45	0.64





3 3) Total decomposition (Term_Loss)

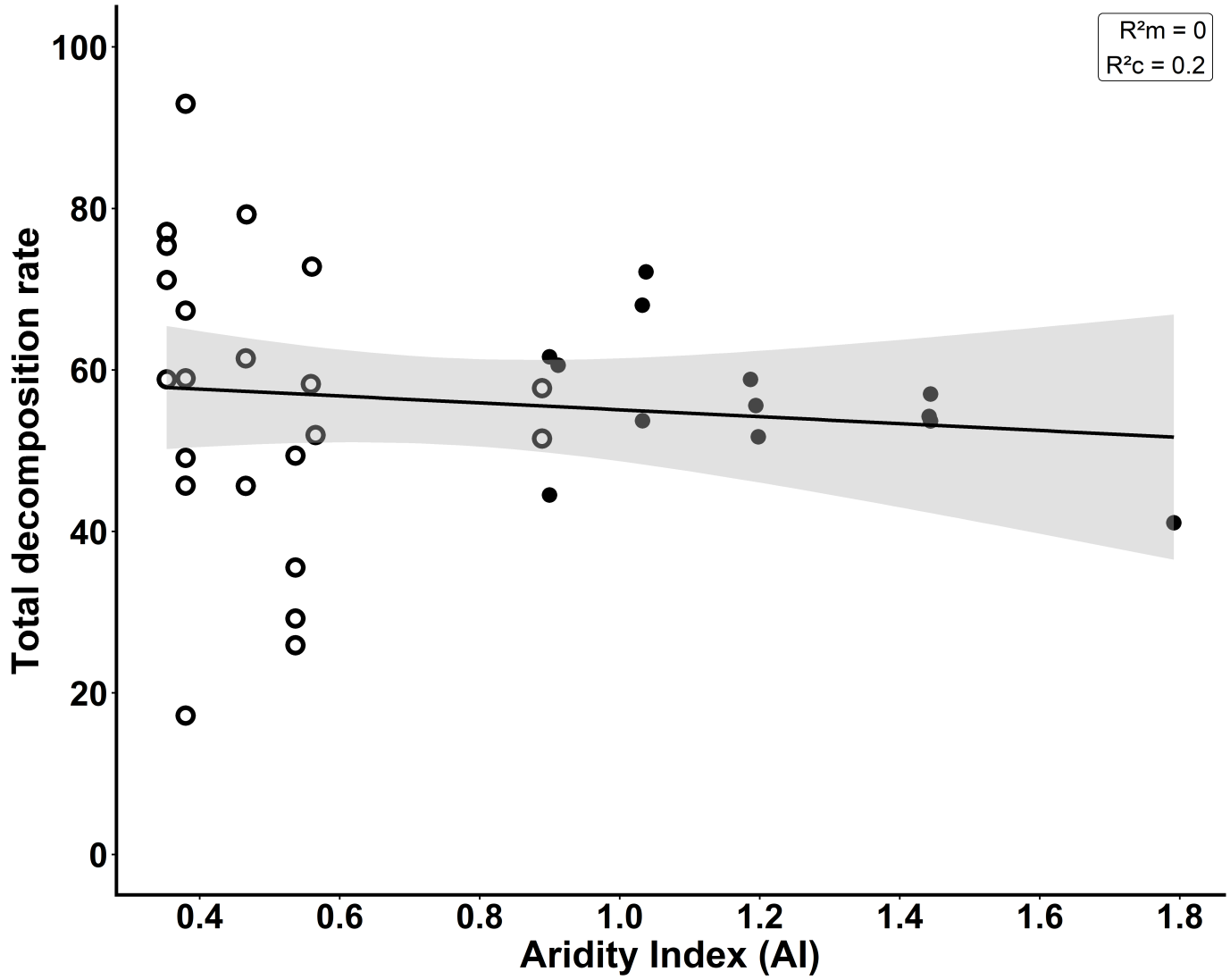
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
56.15	NA	NA	NA	NA	NA	3	-142.40	295.78	0.00	0.32
16.48	NA	1.59	NA	NA	NA	4	-140.49	296.95	1.17	0.18
6.96	-10.39	2.30	NA	NA	NA	5	-136.68	297.95	2.17	0.11
60.52	-5.53	NA	NA	NA	NA	4	-139.16	297.96	2.18	0.11
55.99	NA	NA	+	NA	NA	4	-139.63	298.41	2.63	0.09
22.31	-21.23	2.28	+	NA	NA	6	-132.96	299.49	3.71	0.05
9.75	NA	1.80	+	NA	NA	5	-137.59	299.58	3.79	0.05
73.97	-15.30	NA	+	NA	NA	5	-135.49	299.70	3.92	0.05
0.97	-13.08	2.72	+	+	NA	7	-128.23	301.57	5.78	0.02
-104.67	NA	6.21	+	NA	+	6	-134.11	302.56	6.78	0.01
93.21	-21.87	-0.43	+	NA	+	7	-129.52	302.78	7.00	0.01
72.85	-14.49	NA	+	+	NA	6	-131.03	302.83	7.05	0.01
40.09	-13.52	1.22	+	+	+	8	-124.81	305.42	9.64	0.00

Selected_model_formula
Term_Loss ~ (1 | Site)

nobs	AIC	BIC	logLik	sigma
35	290.79	295.46	-142.4	13.94

term	estimate	std_error	df	t_value
(Intercept)	56.15	3.31	8.75	16.98

R2m	R2c
0	0.2



4 4) Termite proportional (Term_Prop)

(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
89.04	-13.80	NA	+	+	NA	6	-114.47	265.48	0.00	0.30
95.87	-19.92	NA	NA	NA	NA	4	-123.76	265.89	0.41	0.25
108.45	-28.99	NA	+	NA	NA	5	-120.02	266.67	1.18	0.17
121.65	-17.84	-1.10	NA	NA	NA	5	-122.24	267.68	2.19	0.10
134.88	-26.96	-1.12	+	NA	NA	6	-118.49	268.53	3.04	0.07
104.33	-14.55	-0.56	+	+	NA	7	-113.37	268.61	3.13	0.06
115.37	-26.54	-0.39	+	NA	+	7	-115.33	271.92	6.44	0.01
18.82	-12.59	2.65	+	+	+	8	-110.23	272.30	6.81	0.01

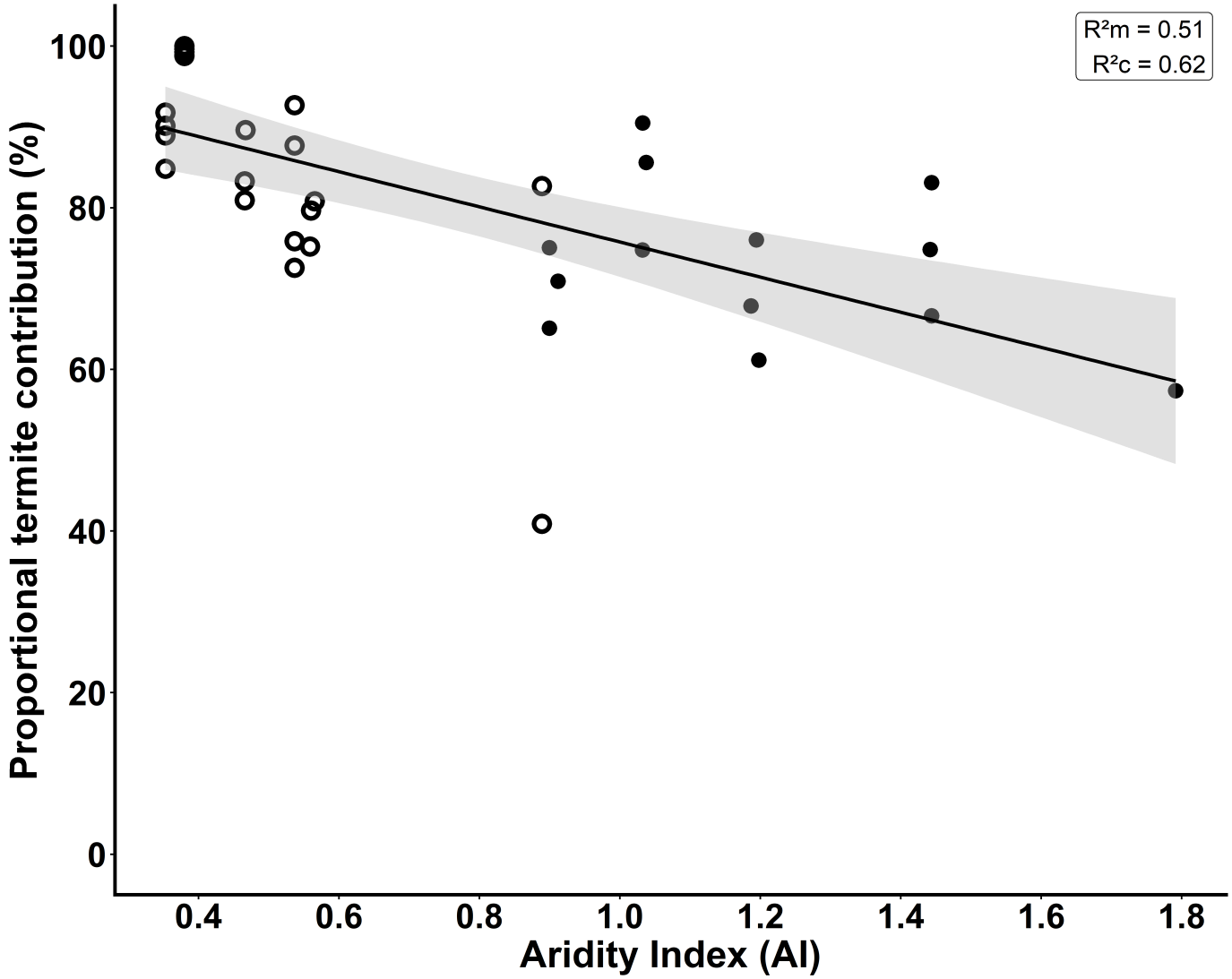
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
79.20	NA	NA	NA	NA	NA	3	-130.57	272.35	6.87	0.01
134.53	NA	-2.21	NA	NA	NA	4	-128.20	272.63	7.15	0.01
76.53	NA	NA	+	NA	NA	4	-127.68	274.05	8.57	0.00
127.37	NA	-2.00	+	NA	NA	5	-125.44	274.72	9.23	0.00
-212.08	NA	11.11	+	NA	+	6	-121.35	275.95	10.47	0.00

Selected_model_formula
Term_Prop ~ AI + VegType + (1 | Site) + AI:VegType

nobs	AIC	BIC	logLik	sigma
35	240.94	250.27	-114.47	8.32

term	estimate	std_error	df	t_value
(Intercept)	89.04	14.01	11.98	6.36
AI	-13.80	11.27	13.42	-1.23
VegTypeSavanna	24.29	17.63	7.59	1.38
AI:VegTypeSavanna	-42.59	20.95	8.30	-2.03

R2m	R2c
0.51	0.62



5 5) Microbe proportional (Microbe_Prop)

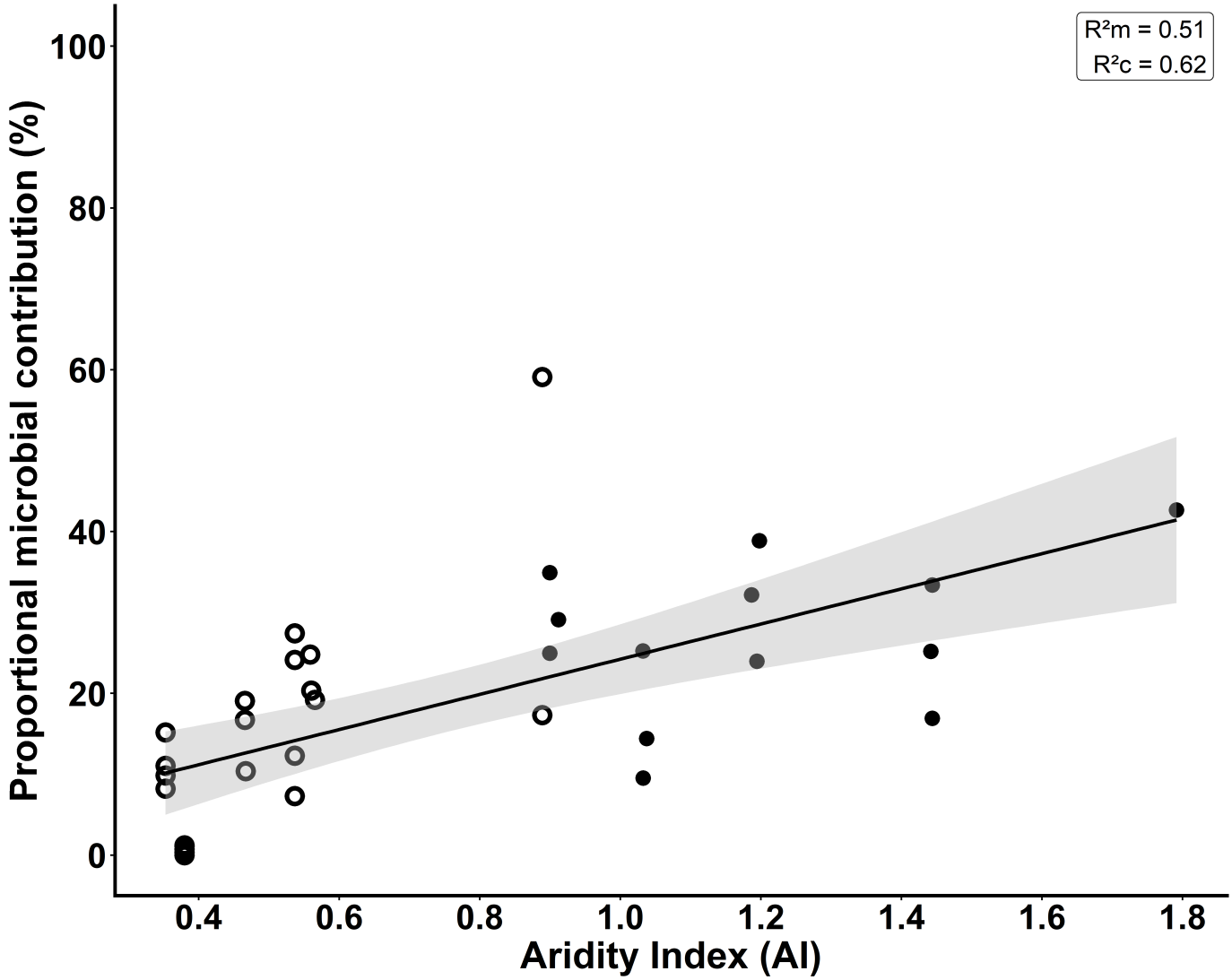
(Intercept)	AI	Avg_Temp	VegType	AI:VegType	Avg_Temp:VegType	df	logLik	AICc	delta	weight
10.96	13.80	NA	+	+	NA	6	-114.47	265.48	0.00	0.30
4.13	19.92	NA	NA	NA	NA	4	-123.76	265.89	0.41	0.25
-8.45	28.99	NA	+	NA	NA	5	-120.02	266.67	1.18	0.17
-21.65	17.84	1.10	NA	NA	NA	5	-122.24	267.68	2.19	0.10
-34.88	26.96	1.12	+	NA	NA	6	-118.49	268.53	3.04	0.07
-4.33	14.55	0.56	+	+	NA	7	-113.37	268.61	3.13	0.06
-15.37	26.54	0.39	+	NA	+	7	-115.33	271.92	6.44	0.01
81.18	12.59	-2.65	+	+	+	8	-110.23	272.30	6.81	0.01
20.80	NA	NA	NA	NA	NA	3	-130.57	272.35	6.87	0.01
-34.53	NA	2.21	NA	NA	NA	4	-128.20	272.63	7.15	0.01
23.47	NA	NA	+	NA	NA	4	-127.68	274.05	8.57	0.00
-27.37	NA	2.00	+	NA	NA	5	-125.44	274.72	9.23	0.00
312.08	NA	-11.11	+	NA	+	6	-121.35	275.95	10.47	0.00

Selected_model_formula
Microbe_Prop ~ AI + VegType + (1 | Site) + AI:VegType

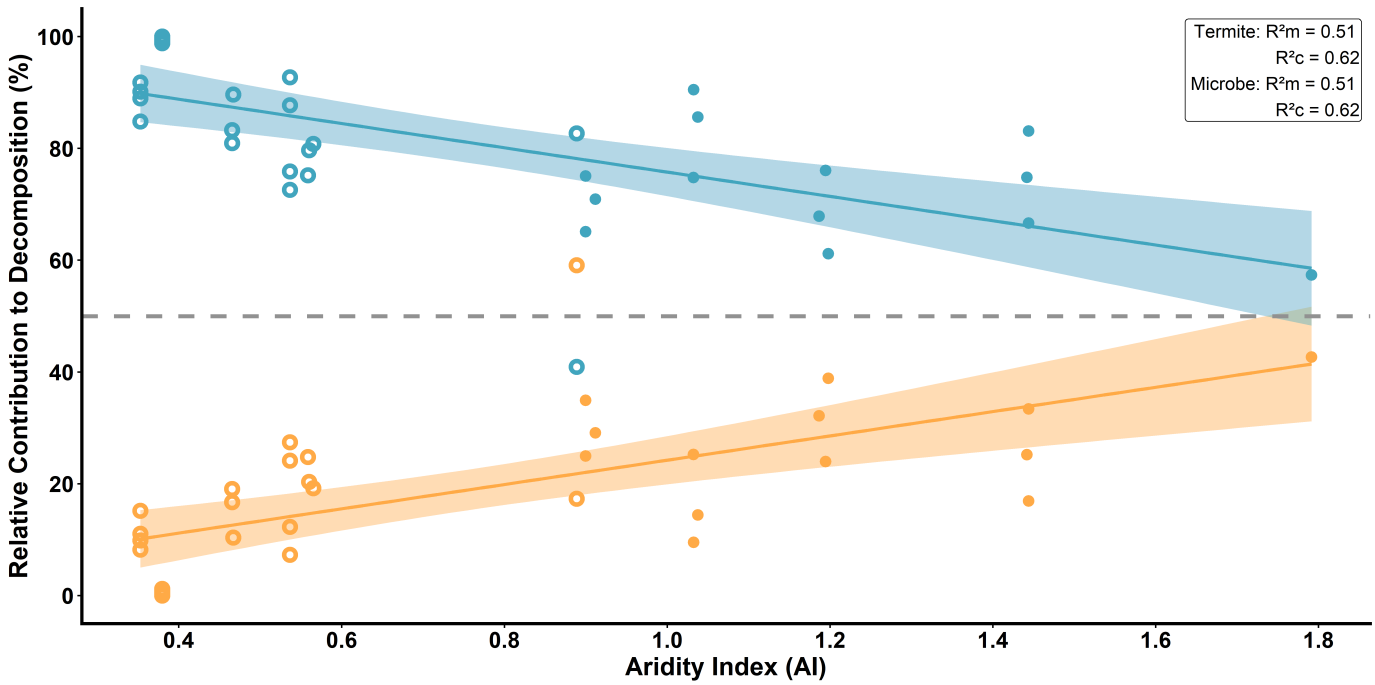
nobs	AIC	BIC	logLik	sigma
35	240.94	250.27	-114.47	8.32

term	estimate	std_error	df	t_value
(Intercept)	10.96	14.01	11.98	0.78
AI	13.80	11.27	13.42	1.23
VegTypeSavanna	-24.29	17.63	7.59	-1.38
AI:VegTypeSavanna	42.59	20.95	8.30	2.03

R2m	R2c
0.51	0.62

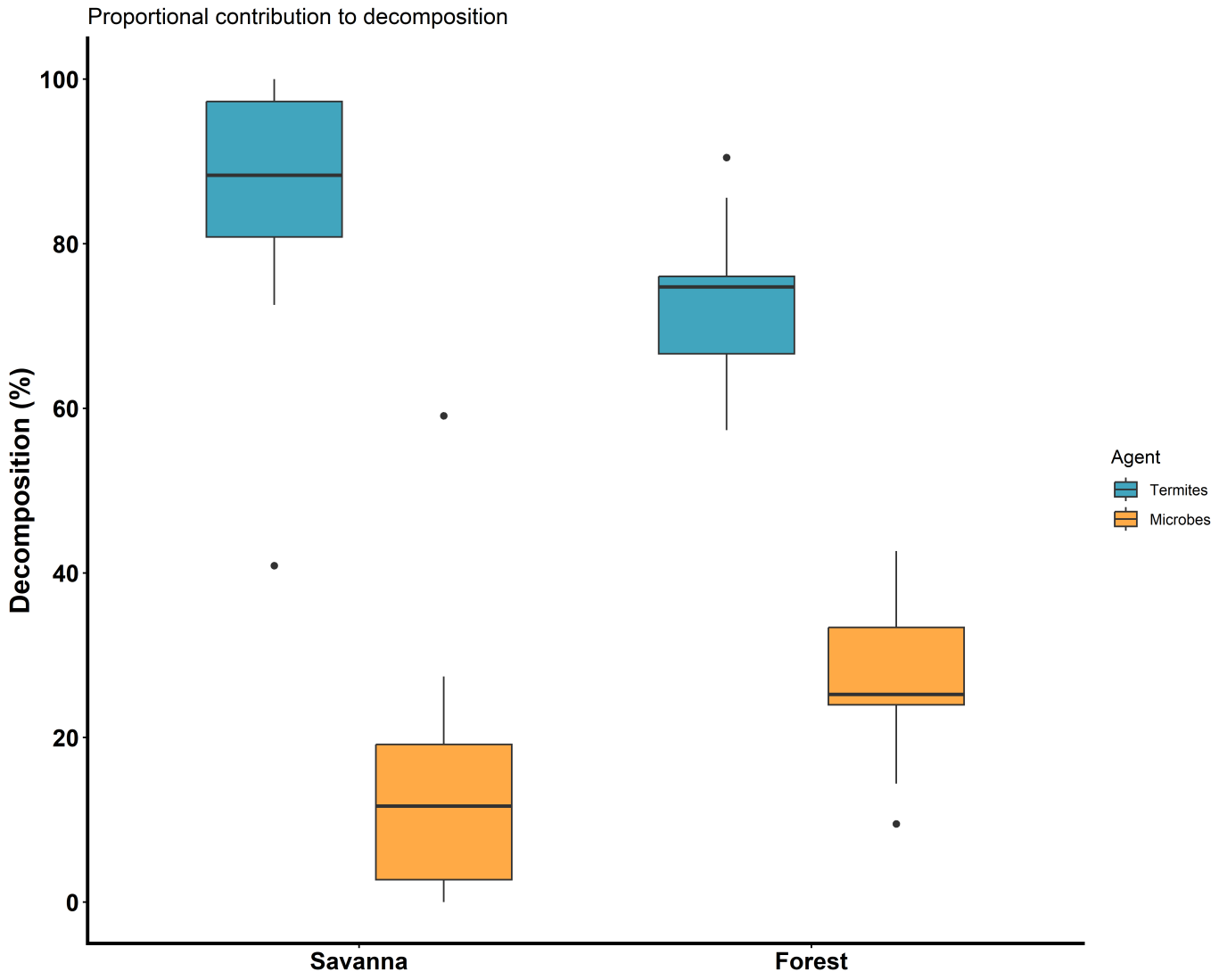


6 6) Combined proportional plot (termite + microbe)

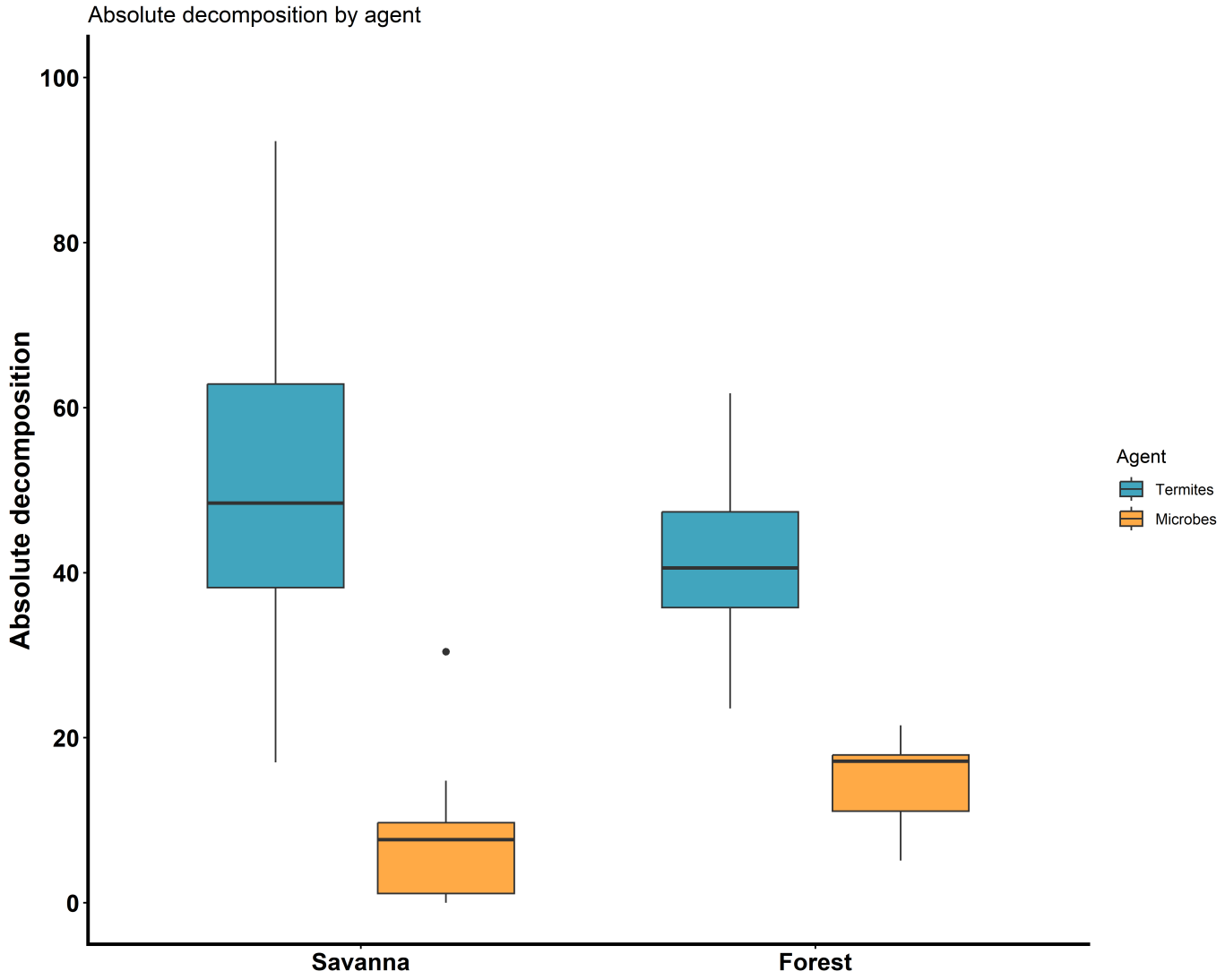


7 7) Boxplots (no points; filled by agent)

7.1 Proportional contributions (0–100)



7.2 Absolute decomposition (0–100)



7.3 Two-way ANOVA: absolute decomposition by

habitat type and decomposition agent

```
library(dplyr)
library(tidyr)
library(car)
library(knitr)

Final_DB <- Final_DB %>%
  mutate(
    VegType = as.factor(VegType)
  )

abs_decomp_long <- Final_DB %>%
  select(VegType, Term_Absolute, Microbe_Absolute) %>%
  pivot_longer(
    cols = c(Term_Absolute, Microbe_Absolute),
    names_to = "Decomposition_type",
    values_to = "Absolute_decomposition"
  ) %>%
  mutate(
    Decomposition_type = factor(
      Decomposition_type,
      levels = c("Term_Absolute", "Microbe_Absolute"),
      labels = c("Termite", "Microbial")
    )
  ) %>%
  filter(
    !is.na(VegType),
    !is.na(Decomposition_type),
    !is.na(Absolute_decomposition)
  )

abs_decomp_aov <- aov(
  Absolute_decomposition ~ VegType * Decomposition_type,
  data = abs_decomp_long
)

abs_decomp_anova <- as.data.frame(car::Anova(abs_decomp_aov, type = 2)) %>%
  tibble::rownames_to_column("term")

knitr::kable(
  abs_decomp_anova,
  digits = 4,
  caption = "Two-way ANOVA for absolute decomposition by habitat type and decomposition agen
t"
)
```

Two-way ANOVA for absolute decomposition by habitat type and decomposition agent

term	Sum Sq	Df	F value	Pr(>F)
VegType	0.4679	1	0.0030	0.9563
Decomposition_type	22355.3700	1	144.5079	0.0000
VegType:Decomposition_type	878.8965	1	5.6813	0.0200

term	Sum Sq	Df	F value	Pr(>F)
Residuals	10210.1957	66	NA	NA

7.4 Mean absolute decomposition by habitat type

```
abs_decomp_summary <- abs_decomp_long %>%
  group_by(VegType, Decomposition_type) %>%
  summarise(
    n = n(),
    mean = mean(Absolute_decomposition, na.rm = TRUE),
    sd = sd(Absolute_decomposition, na.rm = TRUE),
    se = sd / sqrt(n),
    .groups = "drop"
  )

knitr::kable(
  abs_decomp_summary,
  digits = 3,
  caption = "Mean absolute decomposition by habitat type and decomposition agent"
)
```

Mean absolute decomposition by habitat type and decomposition agent

VegType	Decomposition_type	n	mean	sd	se
Forest	Termite	13	41.443	9.610	2.665
Forest	Microbial	13	14.921	4.809	1.334
Savanna	Termite	22	48.607	19.306	4.116
Savanna	Microbial	22	7.418	6.892	1.469